

KCPC-CS111 Research References

Week 2: Investigation Techniques, Extremal Principles, and Tilings

King's Competitive Programming Club
Research Curator: Kutay Bozkurt

This document compiles foundational research papers that underpin the heuristics and problem-solving techniques introduced in Workshop 2. These papers explore the extreme principle in discrete geometry, combinatorial invariants in polyomino tilings, and non-standard number representations.

I. The Extreme Principle and Discrete Geometry

[1] **Sylvester, J. J. (1893) & Gallai, T. (1944)**

Mathematical Question 11851. Educational Times, 59, 98 (1893).

Solution to Problem 4065. American Mathematical Monthly, 51(3), 169–171 (1944). <https://doi.org/10.2307/2304247>

Why referenced: The origin and first published affirmative resolution of the Sylvester-Gallai theorem. Sylvester posed the question of whether a non-collinear finite set of points in the plane must determine an ordinary line (a line containing exactly two points). Tibor Gallai answered in the affirmative using an extremal configuration.

[2] **Kelly, L. M. / Coxeter, H. S. M. (1948)**

A problem of collinear points. American Mathematical Monthly, 55(1), 26–28. <https://doi.org/10.2307/2304500>

Why referenced: Presents L. M. Kelly's celebrated one-paragraph proof of the Sylvester-Gallai theorem using the minimal positive perpendicular distance from a point to a connecting line. This elegant extremal argument is the direct template for the two-color problem solved in Workshop 2.

II. Polyominoes and Tiling Invariants

[3] **Golomb, S. W. (1954)**

Checker boards and polyominoes. American Mathematical Monthly, 61(10), 675–682. <https://doi.org/10.2307/2307321>

Why referenced: Introduced polyominoes into mainstream mathematical literature and established the systematic use of coloring invariants to prove the impossibility of covering boards with missing cells (the mutilated chessboard problem).

[4] Klarner, D. A. (1969)

Packing a rectangle with congruent n -ominoes. Journal of Combinatorial Theory, 7(2), 107–115.
[https://doi.org/10.1016/S0021-9800\(69\)80044-X](https://doi.org/10.1016/S0021-9800(69)80044-X)

Why referenced: Extends coloring arguments to higher-order polyominoes and establishes necessary and sufficient conditions for rectangular grid packings.

III. Number Representations and Fibonacci Bases

[5] Zeckendorf, E. (1972)

Représentation des nombres naturels par une somme de nombres de Fibonacci ou de nombres de Lucas. Bulletin de la Société Royale des Sciences de Liège, 41, 179–182.

Why referenced: Establishes Zeckendorf’s theorem: every positive integer can be uniquely represented as the sum of one or more distinct, non-consecutive Fibonacci numbers. This canonical decomposition guarantees an optimal, greedy decomposition in ≤ 86 steps for integers up to 10^{18} , directly providing the algorithm for ARC 122 C (Calculator).

[6] Lekkerkerker, C. G. (1952)

Voorstelling van natuurlijke getallen door een som van getallen van Fibonacci. Simon Stevin, 29, 190–195.

Why referenced: Proves that the average number of Fibonacci numbers needed to represent an integer n is $\frac{n}{\phi^2+1} \approx 0.276 \log_{\phi} n$, establishing rigorous bounds on the operation counts of Fibonacci-based algorithms.